

Review

Association of bisphenol A exposure with in vitro fertilization outcomes: A meta-analysis and systematic review

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ABSTRACT

Background: The impact of bisphenol A (BPA) on in vitro fertilization (IVF) and intracytoplasmic sperm injection (ICSI) outcomes remains unclear.

Methods: This systematic review and meta-analysis evaluated studies from Pubmed, Web of Science, EMBASE, and Cochrane Library (up to April 19, 2024), analyzing data using regression coefficient (β) and their 95 % confidence intervals (CIs).

Results: A total of 478 references were identified and 14 studies with different IVF/ICSI outcomes were included in the final meta-analysis. The included studies reported that the median or geometric mean (GM) of specific gravity (SG) or creatinine (Cr)-adjusted maternal urinary BPA ranged from 0.063 to 2.61 ng/ml. The meta-analysis results indicated a significant negative correlation between urinary, serum, follicular fluid (FF), and semen BPA exposure and several key IVF/ICSI outcomes, particularly in terms of normal fertilization rates (β : -0.05 ; 95 % CI: -0.07 , -0.03) and the number of high-quality embryos (β : -0.05 ; 95 % CI: -0.09 , -0.01). In the subgroup analysis based on BPA exposure level, higher BPA exposure was significantly associated with decreased normal fertilization rate in IVF/ICSI, particularly in populations with BPA median or GM concentrations above 1.55 ng/ml (β : -0.19 ; 95 % CI: -0.27 , -0.11). However, no significant association was observed between BPA exposure and clinical pregnancy rates, blastocyst formation, or implantation success rates, which may be attributed to the limited number of studies and variability in study design and populations.

Conclusion: This meta-analysis suggests that BPA, a common environmental pollutant, may reduce reproductive success during IVF/ICSI by disrupting endocrine function and compromising gamete quality and early embryo development. However, due to study limitations, further research is needed to confirm our findings and explore the mechanisms underlying reproductive toxicity.

1. Introduction

Bisphenol A (BPA), a common industrial chemical in plastics and packaging, is widely produced and found in human tissues, reflecting its pervasive exposure [1–3]. Following BPA exposure, the compound can be found in human fluids and tissues such as colostrum, breast milk, urine, semen, follicular fluid (FF), and blood [4]. Research has shown that BPA is linked to a range of human diseases, including obesity,

diabetes, cardiovascular diseases, and certain cancers, posing a significant health risk [5–7].

BPA is a recognized xenoestrogen and has estrogen and anti-androgen properties due to its binding affinity with estrogen and androgen receptors [8]. The effects of BPA on reproductive health in female mice extends to the third-generation offspring, despite no direct exposure [9]. Considering the environmental BPA exposure and its ubiquitous presence in human fluids and tissues, BPA imposes an

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increasing impact on the reproductive system. In vitro fertilization (IVF) or intracytoplasmic sperm injection (ICSI) is one of the most effective assisted reproductive technologies, which can be applied to treat infertility, affecting 16 % of couples worldwide [10]. However, despite the significant advances in IVF in recent years, the procedure still faces a high failure rate. BPA exposure could be a potential factor contributing to IVF failure. However, current studies designed to identify BPA's impacts on IVF outcomes have reached controversial conclusions. Many studies have demonstrated that BPA exposure is associated with negative IVF outcomes, including implantation rate, oocyte development, embryo quality, and clinical pregnancy [11–14]; in contrast, other studies reported that BPA exposure had no significant impact on IVF/ICSI outcomes [15].

Therefore, this meta-analysis systematically reviewed the existing literature to clarify the association between prenatal BPA exposure and IVF/ICSI outcomes. To explore potential heterogeneity, subgroup analyses were conducted based on BPA exposure levels, urinary dilution adjustment methods, sample types, and countries.

2. Methods

2.1. Protocol and registry

This systematic review adhered to PRISMA guidelines and was conducted based on the previously published protocol [16]. The review strictly followed the outlined plan, with additional expansions where necessary. All modifications were thoroughly described in the article.

2.2. Search strategy

Relevant studies published before July 2024 were identified from PubMed, Web of Science, EMBASE, and Cochrane Library, with no language restrictions (Table S1). In addition, references were manually searched, and authors were contacted for original data to minimize missing information. Any discrepancies in study selection between XL-C and MK-Y were resolved by consensus.

2.3. Study screening and eligibility criteria

All the literature was managed using Endnote software. First, the duplicates were automatically removed using Endnote and then manually removed by comparing authors, publication dates, and titles. Subsequently, articles were screened by title, abstract, and full text. Inclusion criteria were: human observational or epidemiological studies (case-control, nested case-control, cross-sectional, cohort) involving IVF/ICSI patients with BPA levels measured in urine, blood, or FF before IVF; outcomes assessed using odds ratios (ORs), relative risks (RRs), or β values with 95 % confidence intervals (CIs). Studies involving donor egg, sperm, or embryo cycles, or lacking data on BPA and IVF/ICSI outcomes, were excluded.

2.4. Data extraction

Data extraction was performed by two independent authors (XL-C and MK-Y), and the information about study year, country, study design, ART type, sample size, gender, age, biological sample, sample time, measurement, dilution adjustment methods, concentration of BPA, covariate adjustment in the model, IVF/ICSI outcomes (retrieved oocytes, mature oocytes, normal fertilization, high-quality embryos, blastocyst formation, high-quality blastocyst, implantation, clinical pregnancy) was extracted as a basic clinical data. In studies not reporting the specific number of subjects for risk analysis, the total number of eligible patients was recorded as the sample size. To demonstrate the association between BPA and IVF/ICSI outcomes, the ORs, RRs, and β values were extracted for further analysis.

2.5. Quality assessment

Quality assessment of the included studies was conducted using the Newcastle-Ottawa Scale (NOS), which evaluates studies based on selection, comparability, and outcome [17]. The NOS consists of three categories with eight items, awarding a total score from 0 to 9 stars: up to 4 stars for Selection, 2 for Comparability, and 3 for Outcome. Studies with 6 or more stars are considered high quality, 3–5 stars as medium quality, and fewer than 3 stars as low quality (Table S2).

2.6. Statistical analysis

Since most studies reported risk estimates as β for IVF/ICSI outcomes, the association between prenatal BPA exposure and these outcomes was assessed by calculating pooled β and 95 % CIs. Additionally, OR and RR can be converted to β coefficients in logistic regression by using the natural logarithm, with $\beta = \ln(\text{OR})$ or approximately $\beta \approx \ln(\text{RR})$ when the baseline risk is small [18,19]. If the studies reported ORs or RRs as risk estimates for IVF/ICSI outcomes, the association between prenatal BPA exposure and IVF/ICSI outcomes was evaluated by calculating the converted pooled β and converted 95 % CI. All pooled results were calculated using the DerSimonian and Laird random-effects model, accounting for both within-study and between-study variability. The summary measures were depicted as forest plots, with the size of the squares representing study-specific β coefficients proportional to their statistical weight, representing the inverse of the variance of the natural logarithm of β . The diamonds represent the pooled β coefficients with their corresponding 95 % CIs. In this meta-analysis, β coefficients represent the estimated change in the IVF/ICSI outcome per unit increase in BPA concentration (e.g., ng/ml). A negative β indicates that higher BPA exposure is associated with a decline in the outcome measure (e.g., fertilization rate or number of oocytes). The strength of the association is assessed by the absolute magnitude of β and its 95 % confidence interval (CI), while statistical significance is determined by whether the 95 % CI excludes zero and by the associated p-value. Statistical heterogeneity was assessed using the I^2 statistic; a random-effects model was used when $I^2 > 50\%$ or $P < 0.05$, otherwise a fixed-effect model was applied. A sensitivity analysis was performed to evaluate the impact of each study on the overall results for clinical pregnancy and high-quality embryos. Publication bias was examined using funnel plots and Begg's test, with the standard error plotted against effect size (β) to assess asymmetry and potential bias. In studies including multiple groups, the results from the different groups were first combined, and a pooled analysis was conducted with the results from other studies. Since the present study focused on the association between BPA exposure and IVF/ICSI outcomes, other groupings were treated as independent samples and combined accordingly. Meta-analysis was performed using the Metan module in STATA version 17 software.

3. Results

3.1. Characteristics of the studies

Fig. 1 illustrates the article screening process based on the inclusion criteria. A total of 478 articles were identified from PubMed, Web of Science, EMBASE, and Cochrane Library up to April 19, 2024. After removing 166 duplicates, 312 titles and abstracts were screened. Of these, 273 studies were excluded (74 reviews, 2 intervention studies, 125 non-clinical studies, 24 non-IVF/ICSI studies, and 49 unrelated studies), leaving 38 for full-text and reference review. After full-text assessment, 24 studies were excluded: 20 for insufficient data and 4 for outcomes not suitable for the meta-analysis. Additionally, the references of the included full-text articles were manually searched, but no eligible studies were found. Finally, 14 studies were included, with 7 studies reporting the retrieved oocytes of IVF/ICSI, 4 reporting the mature oocytes of IVF/ICSI, 8 reporting the normal fertilization in IVF/

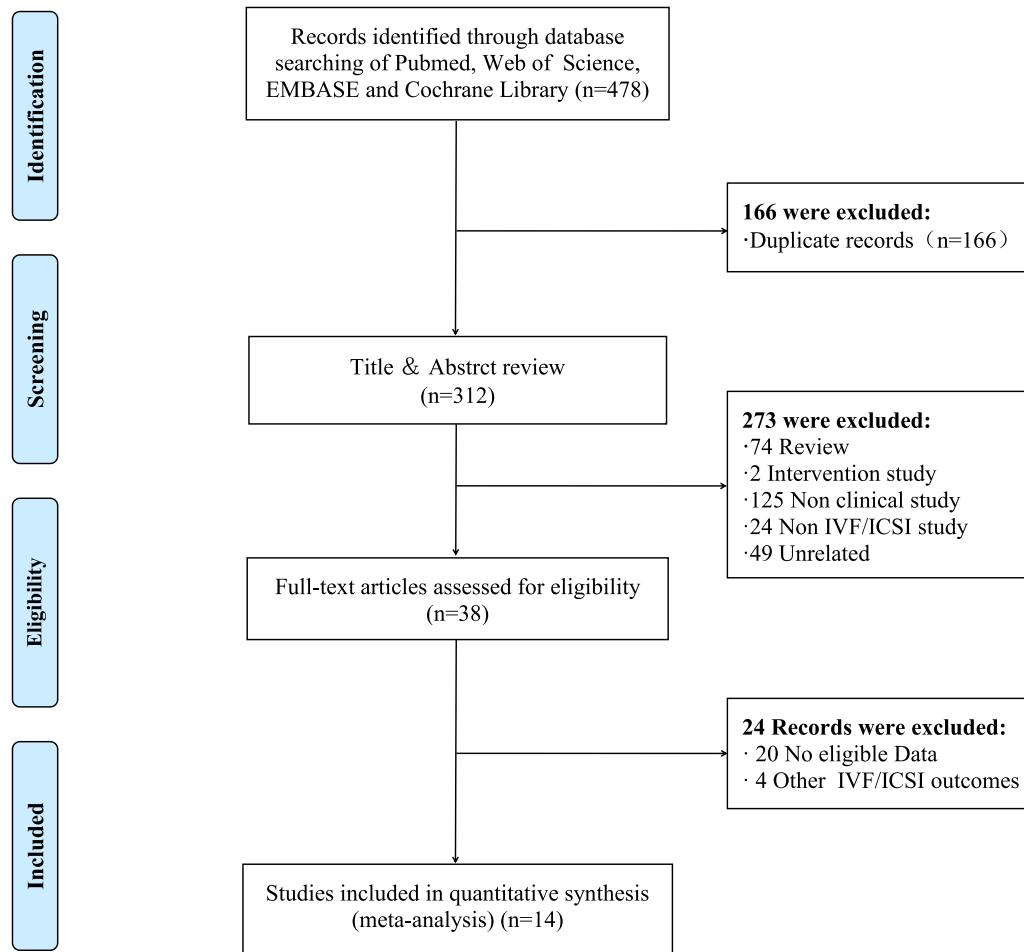


Fig. 1. The PRISMA flow diagram. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

ICSI, 5 reporting the high-quality embryos of IVF/ICSI, 2 reporting the blastocyst formation of IVF/ICSI, 2 reporting the high-quality blastocysts of IVF/ICSI, 5 reporting the implantation of IVF/ICSI, and 5 reporting the clinical pregnancy of IVF/ICSI (Table S3).

Table 1 presents the baseline characteristics of the included studies. A total of 14 studies were analyzed, consisting of 13 prospective cohort studies and 1 cross-sectional study, focusing on the pooled β for IVF/ICSI outcomes. These studies, published between 2010 and 2024, involved 1985 women and 782 men, and examined the relationship between BPA exposure in urine, blood, FF, or semen and IVF/ICSI outcomes. The studies examining the association between BPA exposure and IVF/ICSI outcomes had sample sizes ranging from 44 to 450, with half including more than 100 participants. Most studies adjusted for covariates such as maternal and paternal age, BMI, smoking, alcohol use, education, income, and infertility diagnosis. Table 1 presents the NOS quality assessment results, with scores ranging from 6 to 9, and an average of 8. Based on these scores, the majority of the studies were classified as high quality.

3.2. Evaluation of BPA exposure assessment during IVF/ICSI

All studies assessed BPA exposure in maternal or paternal samples collected on the day of oocyte retrieval, as shown in Table 1. Most of these studies focused on urinary BPA levels in either the mother or father. Only four studies, including those from Bloom et al., Fujimoto et al., Kim et al., and Ozelci et al., examined BPA exposure in maternal or paternal plasma samples [12,15,20,21]. In addition, only two studies, including those from Kim et al. and Mina et al., assessed BPA exposure in maternal follicular fluid samples, and only one study by Kim et al.

evaluated BPA exposure in paternal semen samples [15,22]. Only one study encompassed both female and male BPA exposure, including all sample types [15]. The included studies yielded a median or geometric mean (GM) of maternal urinary BPA measured by MS or HPLC ranging from 0.063 to 2.61 ng/ml.

Nonetheless, the studies that reported the association between BPA exposure and IVF/ICSI outcomes measured BPA in urine, FF, serum, or semen samples from different IVF/ICSI procedure periods. However, the meta-analysis only included studies in which BPA was measured on the day of oocyte retrieval. Eight studies used urinary specific gravity (SG) to adjust for urinary BPA concentrations [11,14,15,23–26] and two studies used urinary creatinine (Cr) [27,28], while others used unadjusted urinary BPA concentrations [12,20–22].

3.3. BPA exposure and IVF/ICSI outcomes

The combined results of studies measuring BPA in urine, serum, FF, and semen samples indicated BPA exposure was significantly associated with reduced normal fertilization rate ($\beta = -0.05$, 95 % CI: -0.07 , -0.03 , $P < 0.001$), showing moderate statistical heterogeneity ($I^2 = 56.90\%$, $P < 0.05$) (Fig. 2A). Moreover, the six studies measuring BPA in urine and the two studies in serum of maternal or paternal origin also demonstrated a similar negative association. The combined β was -0.05 (95 % CI: -0.07 , -0.02) and -0.09 (95 % CI: -0.17 , -0.02), with low ($I^2 = 49.60\%$, $P > 0.05$) and high ($I^2 = 80.90\%$, $P < 0.05$) heterogeneity in urine and serum samples, respectively (Fig. 2A). However, no significant association was found in FF and semen samples, which may be attributed to the small number of studies included (1 and 2) (Fig. 2A). The subgroup analysis by BPA level revealed a significant association

Table 1
Main characteristics of included studies.

Study	Year	Country	Study design	ART type	Sample size (N)	Gender	Age	Biological sample	Sample time	Measurement	Dilution adjustment	Concentration of BPA	Covariate adjustment in the model	IVF outcomes	NOS score
Radwan et al.	2020	Poland	Prospective cohort study	ICSI	450	Female	31.28 ± 3.52 ¹	Urinary	Before oocyte retrieval	GC-ITMS	SG-adjusted	1.62 (0.95, 2.52) ²	BMI, age, smoking, infertility diagnosis	Retrieved oocytes, mature oocytes, normal fertilization, high quality embryos, implantation, clinical pregnancy	9
Mina et al.	2022	Greece	Cross-sectional study	ICSI	45 48	Female	PCOS: 35 (34, 38) ² TFI: 38 (37, 41) ²	FF	Day of oocyte retrieval	GC-MS	Unadjusted	0.50 (0.14, 1.15) ² 1.13 (0.40, 2.05) ²	Age, BMI, FSH	Retrieved oocytes, normal fertilization	7
Ehrlich et al.	2012	America	Prospective cohort study	ICSI/IVF	174	Female	35.6 ± 4.0 ¹	Urinary	Day of oocyte retrieval	SPE-HPLC-MS/MS	SG-adjusted	2.32 (1.60,3.77) ²	Age, BMI, FSH	Retrieved oocytes, mature oocytes, normal fertilization	8
Minguez-Alarcon et al.	2015	America	Prospective cohort study	ICSI/IVF	256	Female	35.0 (32.5, 39.0) ²	Urinary	Day of oocyte retrieval	SPE-HPLC-MS/MS	SG-adjusted	1.38 (0.97, 2.24) ²	Age, BMI, race, smoking, infertility diagnosis	Implantation, clinical pregnancy	9
Ehrlich et al.	2012	America	Prospective cohort study	IVF	137	Female	35.8 ± 4.0 ¹	Urinary	Day of oocyte retrieval	SPE-HPLC-MS/MS	SG-adjusted	2.33 (1.69, 3.79) ²	Age, day of embryo transfer, IVF protocol	Implantation	9
Kim et al.	2021	Korea	Prospective cohort study	ICSI	146 146	Female Male	35 (31, 38) ² 36 (32, 40) ²	Urinary Serum Urinary Serum Semen	Day of oocyte retrieval	LC-MS/MS	SG-adjusted Unadjusted Unadjusted SG-adjusted Unadjusted Unadjusted	0.66 (0.27,5.13) ² 0.118 (0.036,0.229) ² 0.063 (0.021,0.147) ² 1.64 (0.33,5.16) ² 0.105 (0.019,0.178) ² 0.068 (<LOD,0.146) ²	Age, BMI, race, AMH	Retrieved oocytes, normal fertilization, high quality embryos, clinical pregnancy Normal fertilization, high quality embryos, clinical pregnancy	9
Yang et al.	2023	China	Prospective cohort study	IVF/ICSI	192 157	Female Male	33.7 ± 4.9 ¹ 35.3 ± 5.4 ¹	Urinary	Day of oocyte retrieval	UPLC-MS/MS	SG-adjusted	0.06 (<LOD, 0.45) ² 0.16 (<LOD, 0.46) ²	Age, BMI, smoking, alcohol drinking, education level, income, infertility diagnosis, insemination technique, parity Age, BMI, smoking, alcohol drinking, education level, income, infertility diagnosis, insemination technique, fathered a pregnancy	Retrieved oocytes, mature oocytes, normal fertilization, high quality embryos, blastocyst formation, high quality blastocyst Normal fertilization, high quality embryos, blastocyst formation, high quality blastocyst	8
Dodge et al.	2015	America	Prospective cohort study	IVF/ICSI	218	Male	36.7 ± 5.1 ¹	Urinary	Day of oocyte retrieval	SPE-HPLC-MS/MS	SG-adjusted	1.8 (1.1,2.5) ²	Age, weight, smoking	Normal fertilization, high quality embryos, implantation	9

(continued on next page)

Table 1 (continued)

Study	Year	Country	Study design	ART type	Sample size (N)	Gender	Age	Biological sample	Sample time	Measurement	Dilution adjustment	Concentration of BPA	Covariate adjustment in the model	IVF outcomes	NOS score
Fujimoto et al.	2011	America	Prospective cohort study	IVF/ICSI	58	Female	35.8 ± 3.9 ¹	Serum	Day of oocyte retrieval	HPLC	Unadjusted	2.53 (0.0–67.4) ³	Age, race, smoking	Mature oocytes, normal fertilization,	7
					37	Male	38.4 ± 4.3 ¹	Serum	Day of oocyte retrieval			0.34 (0.0–22.7) ³		Normal fertilization	
Knez et al.	2014	Slovenia	Prospective cohort study	IVF/ICSI	149	Male	34.05 ± 4.76 ¹	Urinary	Day of oocyte retrieval	GC-MS	Cr-adjusted	1.4 (0.69, 2.63) ²	Age, BMI, IVF/ICSI	Normal fertilization, high quality embryos, blastocyst formation, high quality blastocyst Retrieved oocytes	7
Mok-Lin et al.	2010	America	Prospective cohort study	IVF	84	Female	35.6 ± 3.9 ¹	Urinary	Day of oocyte retrieval	SPE-HPLC-MS/MS	SG-adjusted	2.61 (1.44 , 4.32) ²	Age, BMI, FSH	Retrieved oocytes	7
Bloom et al.	2011	America	Prospective cohort study	IVF	44	Female	35.80 ± 4.08 ¹	Serum	Day of oocyte retrieval	HPLC	Unadjusted	2.53 (0.52 , 6.31) ²	Race, smoking, AFC	Retrieved oocytes	6
Shen et al.	2020	China	Prospective cohort study	IVF	351	Female	31 (28,34) ²	Urinary	Day of oocyte retrieval	HPLC-ESI-MS/MS	Cr-adjusted	0.72 (0.246, 2.021) ²	Age, BMI, FSH, E2, AFC	Implantation, clinical pregnancy	9
Ozelci et al.	2024	Turkey	Prospective cohort study	IVF/ICSI	75	Male	34.9 ± 6.71	Serum	Day of oocyte retrieval	ELISA	Unadjusted	27.7 ± 24.81	Sperm concentration, sperm motility, total progressive mobile sperm count	Clinical pregnancy	8

Note: ¹Mean±SD, ²median (Q1,Q3), ³median (range)

ART: assisted reproductive technology; ICSI: intracytoplasmic sperm injection; IVF: in vitro fertilization; FF: follicular Fluid; GC-ITMS: gas chromatography ion-tap mass spectrometry; GC-MS: GC-mass spectrometry; SPE-HPLC-MS/MS: online solid-phase extraction coupled with isotope dilution-high-performance liquid chromatography-tandem MS; LC-MS/MS: liquid chromatography coupled with tandem MS; UPLC-MS/MS: ultra-performance liquid chromatography coupled with quadrupole MS; HPLC-ESI-MS/MS: HPLC-electron spray ionization-tandem MS; SG: specific Gravity; ELISA: enzyme-linked immunosorbent assay; BPA: bisphenol A; Cr: creatinine; AFC: antral follicle count; NOS: Newcastle-Ottawa Scale; BMI: body mass index; AMH: anti-Mullerian hormone; FSH: follicle stimulating hormone; E2: estradiol.

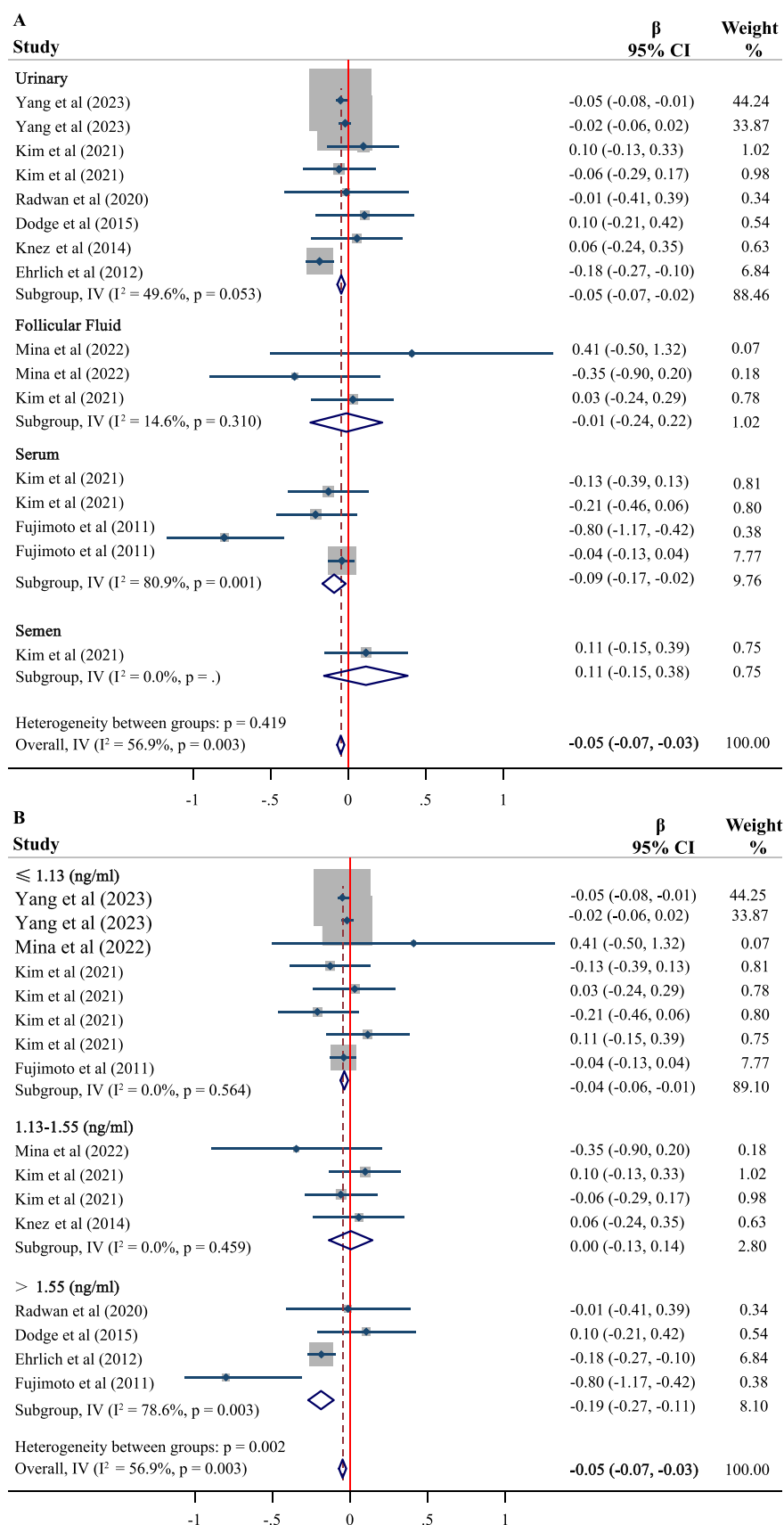


Fig. 2. Forest plots of BPA exposure and normal fertilization of IVF/ICSI Forest plot of the association between BPA exposure and normal fertilization of IVF/ICSI based on biological sample type (A) and BPA level (B). The diamonds indicate study-specific coefficients (β) (size of the diamonds reflects the study-specific statistical weight); the horizontal lines indicate 95 % CIs, and the diamond indicates the summary β with the related 95 % CIs. CI, confidence interval; IVF, in vitro fertilization; ICSI, intracytoplasmic sperm injection.

between higher BPA exposure and reduced normal fertilization in IVF/ICSI, particularly in populations with BPA median concentrations above 1.55 ng/ml (β : -0.19 ; 95 % CI: -0.27 , -0.11), showing high heterogeneity ($I^2 = 78.60$ %, $P < 0.05$) (Fig. 2B). Additionally, the combined results indicated a decrease in high-quality embryos of IVF/ICSI associated with BPA exposure (β : -0.05 ; 95 % CI: -0.09 , -0.01), with low heterogeneity ($I^2 = 31.80$ %, $P > 0.05$) (Fig. 3A). The combined β across five studies measuring BPA in urine samples was -0.05 (95 % CI: -0.10 , -0.01), also showing low heterogeneity ($I^2 = 52.40$ %, $P > 0.05$) (Fig. 3A). However, serum, FF, and semen samples showed no significant association, likely owing to the small number of studies (2, 1, and 1) (Fig. 3A). The subgroup analysis by BPA levels showed a significant correlation between decreased high-quality embryos of IVF/ICSI in populations and BPA concentrations below 1.16 ng/ml (β : -0.06 ; 95 % CI: -0.10 , -0.01), demonstrating low heterogeneity ($I^2 = 0.00$ %, $P > 0.05$) (Fig. 3B). Furthermore, combined results from studies measuring BPA in various samples showed no significant association with clinical pregnancy in IVF/ICSI (β : 0.01 ; 95 % CI: -0.06 , 0.08), showing low heterogeneity ($I^2 = 13.20$ %, $P > 0.05$) (Fig. 4A). Subgroup analysis by sample type and BPA levels also showed no significant link between BPA exposure and clinical pregnancy in IVF/ICSI (Fig. 4B).

Interestingly, the combined results of studies measuring BPA in urine, serum, and FF also indicated that BPA exposure was linked to a decrease in retrieved oocytes and mature oocytes in IVF/ICSI (β : -0.06 ; 95 % CI: -0.08 , -0.03) and (β : -0.04 ; 95 % CI: -0.07 , -0.02), showing low and high statistical heterogeneity ($I^2 = 32.50$ %, $P > 0.05$) and ($I^2 = 85.10$ %, $P < 0.05$), respectively (Fig. S1 and 2 A). The subgroup analysis based on BPA levels also indicated a concentration-dependent decrease in retrieved oocytes and mature oocytes of IVF/ICSI, consistent with the aforementioned results based on sample type (Fig. S1 and 2B). In contrast, the combined results of studies measuring BPA in urine, serum, semen, and FF showed no significant association between BPA exposure (regardless of sample type or BPA level) and other IVF/ICSI outcomes such as blastocyst formation, high-quality blastocysts, and implantation (β : -0.02 ; 95 % CI: -0.06 , 0.02), (β : -0.03 ; 95 % CI: -0.09 , 0.03), and (β : 0.46 ; 95 % CI: 0.42 , 0.51), respectively (Fig. S3-5A and B). Notably, the number of included studies for these indicators was relatively small; hence, these results require further validation.

Subgroup analyses were conducted according to the predetermined factors of BPA levels, gender, country, and urinary dilution adjustment to further explore the sources of heterogeneity (Table 2). The results revealed a significant association between decreased retrieved oocytes, mature oocytes, and normal fertilization in IVF/ICSI and BPA exposure in China and America. In contrast, no significant association was observed for high-quality embryos, blastocyst formation, high-quality blastocysts, implantation, and clinical pregnancy of IVF/ICSI with BPA exposure in different countries (Table 2). The results of the subgroup analyses based on different urinary dilution adjustment methods are displayed in Table 2. The studies that adjusted BPA levels for specific gravity (SG) demonstrated a significant association between decreased retrieved oocytes, mature oocytes, and high-quality embryos and normal fertilization rates in IVF/ICSI. Conversely, no significant associations were found in studies where BPA concentration was either unadjusted or adjusted for creatinine (Cr). Furthermore, no significant associations were observed between BPA exposure and blastocyst formation, high-quality blastocysts, implantation, or clinical pregnancy rates in IVF/ICSI across the different urinary dilution adjustment methods (Table 2).

3.4. Sensitivity analysis and publication bias

A sensitivity analysis was conducted using leave-one-out forest plots with random- or fixed-effect models to assess the stability of our results. The analysis showed that excluding any single study had little impact on the pooled β for BPA exposure and its association with normal fertilization, high-quality embryos, or clinical pregnancy in IVF/ICSI,

confirming the robustness of our findings (Table S4-6). Additionally, Egger's regression test and funnel plots indicated no significant publication bias in the association between BPA exposure and high-quality embryos or clinical pregnancy in IVF/ICSI (Fig. 5). These findings suggest that our conclusions are reliable and not influenced by individual studies. Furthermore, risk of bias was evaluated using the NOS for observational studies. Studies with scores of 6 or higher were classified as moderate to high quality, while those scoring below 6 were considered at high risk of bias. To mitigate this, sensitivity analyses were performed excluding high-risk studies, ensuring the robustness of the results.

4. Discussion

This meta-analysis and systematic review offers a thorough evaluation of the relationship between BPA exposure in various samples and IVF outcomes. The results suggest a potential negative impact of BPA on several IVF outcomes, particularly normal fertilization and embryo quality. Despite variability in sample types, exposure levels, assessment periods, and dilution adjustments across studies, the associations were generally consistent in the subgroup analyses. These findings indicate that BPA, a widespread environmental contaminant, may jeopardize reproductive success in individuals undergoing IVF.

The primary finding of this study is the consistent association between BPA exposure and the reduction in normal fertilization rates, as evidenced by the significant negative pooled β across various studies. Subgroup analyses further supported the detrimental effect of BPA on fertilization, revealing a dose-response relationship, with higher BPA concentrations correlating with more pronounced declines in fertilization success. These findings aligned with previous research suggesting that BPA disrupts endocrine function, potentially impairing gamete quality and early embryonic development [9,11,15,20,29]. In addition to fertilization, this study also highlighted the negative impact of BPA on embryo quality. The meta-analysis demonstrated a significant association between higher BPA levels and a reduction in the number of high-quality embryos, a critical factor for successful IVF outcomes. This relationship was particularly evident in studies where BPA levels were adjusted for urinary specific gravity, emphasizing the importance of accurate exposure assessment in understanding BPA's effects on reproductive health [12,30-32]. However, the study revealed no significant association between BPA exposure and clinical pregnancy rates, blastocyst formation, or implantation success. The lack of significant findings in these areas may be attributed to the limited number of studies and the potential variability in study designs, populations, and exposure assessment methods. Additionally, clinical pregnancy is influenced by a myriad of factors beyond fertilization and early embryonic development, which might explain the absence of a clear link with BPA exposure in the current analysis [33]. BPA alternatives are also classified as reproductive and developmental toxicants and are suspected endocrine disruptors [34]. Exposure to substitutes such as BPS and BPF has been shown to directly impair sperm function, leading to compromised embryo quality [35]. Similarly, exposure to BPB and BPM disrupts meiotic maturation and impairs oocyte quality through mechanisms including DNA damage, altered α -tubulin acetylation, and disrupted epigenetic modifications [36,37]. Moreover, our findings aligned with the review by Machtinger et al. in animal and human model studies, indicating that BPA exposure may adversely affect oocyte growth and IVF outcomes; nonetheless, the present study integrates a larger body of human research data [38]. The majority of studies employed urinary BPA concentrations as a biomarker, adjusting for specific gravity or creatinine levels to account for sample dilution. Our findings indicate that studies adjusting for specific gravity yield stronger associations with adverse IVF outcomes compared to creatinine adjustments, emphasizing the need for standardized BPA assessment methods across studies.

Animal studies have revealed that exposure to certain concentrations of BPA during pregnancy can lead to abnormal fetal and placental

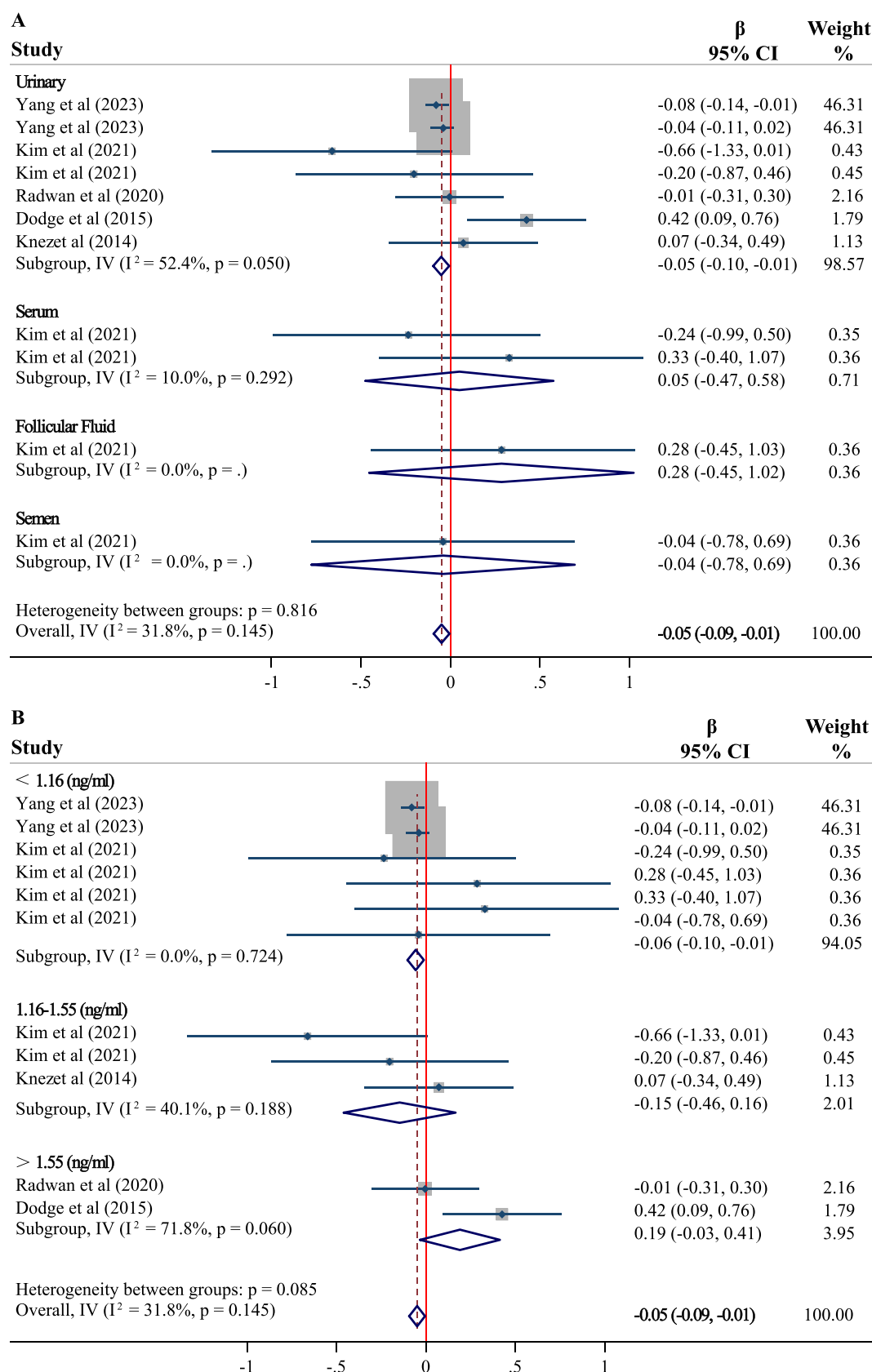


Fig. 3. Forest plots of BPA exposure and high quality embryos of IVF/ICSI Forest plot of the association between BPA exposure and high quality embryos of IVF/ICSI based on biological sample type (A) and BPA level (B). The diamonds indicate study-specific coefficients (β) (size of the diamonds reflects the study-specific statistical weight); the horizontal lines indicate 95 % CIs, and the diamond indicates the summary β with the related 95 % CIs. CI, confidence interval; IVF, in vitro fertilization; ICSI, intracytoplasmic sperm injection.

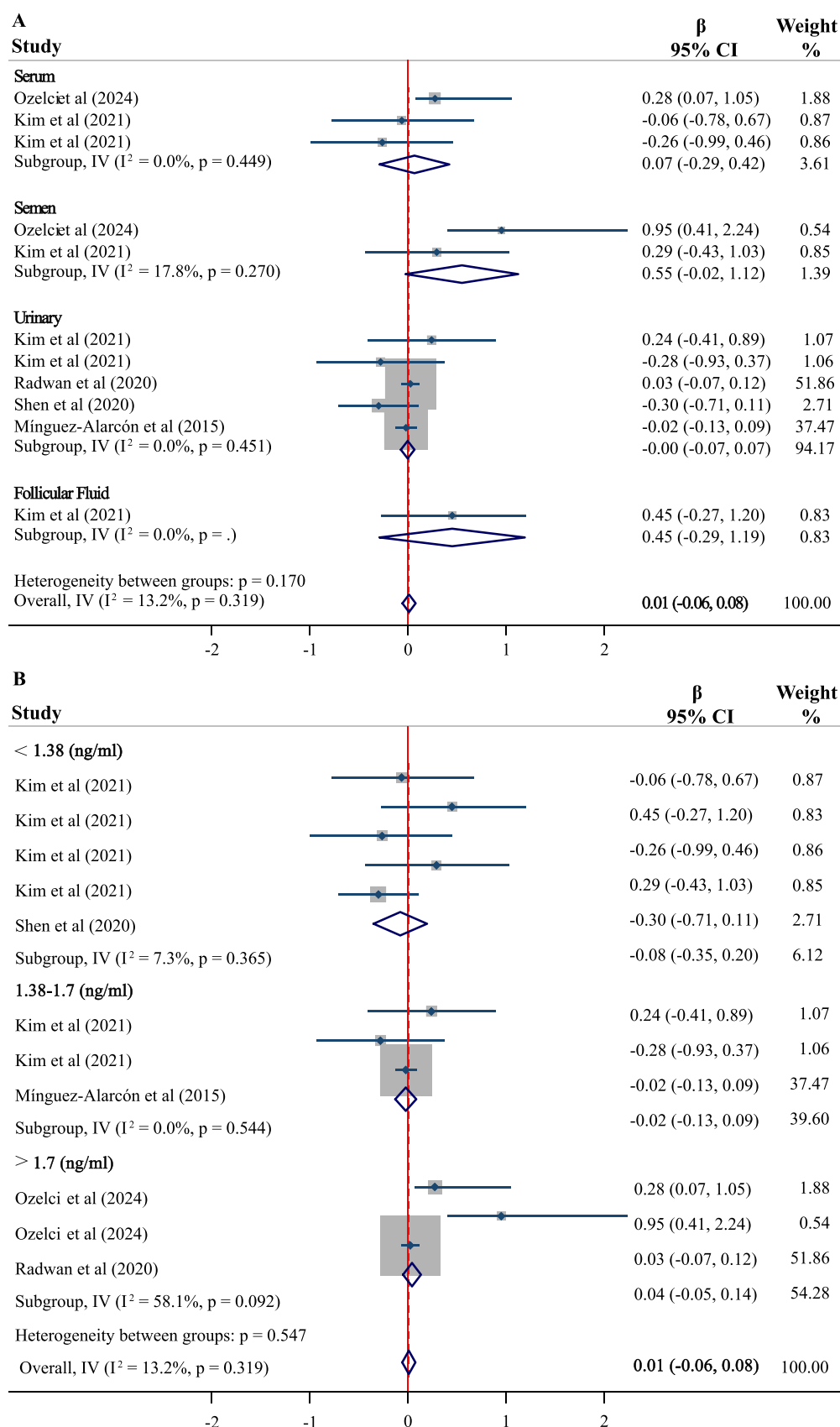


Fig. 4. Forest plots of BPA exposure and clinical pregnancy of IVF/ICSI Forest plot of the association between BPA exposure and clinical pregnancy of IVF/ICSI based on biological sample type (A) and BPA level (B). The diamonds indicate study-specific coefficients (β) (size of the diamonds reflects the study-specific statistical weight); the horizontal lines indicate 95 % CIs, and the diamond indicates the summary β with the related 95 % CIs. CI, confidence interval; IVF, in vitro fertilization; ICSI, intracytoplasmic sperm injection.

Table 2
Main results in the total and subgroup analysis.

Study groups	Retrieved oocytes			Mature oocytes		Blastocyst formation		High quality blastocysts		Implantation	
Overall (β (95 % CI))	−0.06 (−0.08, −0.03)			−0.04 (−0.07, −0.02)		−0.02 (−0.06, 0.02)		−0.03 (−0.09, 0.03)		0.46 (0.42, 0.51)	
Subgroup analysis of BPA exposure and retrieved oocytes of IVF/ICSI											
By urinary dilution adjustment method (β (95 % CI)		By BPA level (β (95 % CI)			By country (β (95 % CI)						
SG-adjusted	Unadjusted	< 1.13 (ng/ml)	1.13–1.70 (n/ ml)	> 1.70 (ng/ml)	China	Greece	Korea	Poland	America		
−0.06 (−0.09, −0.03)	−0.06 (−0.16, 0.04)	−0.04 (−0.07, −0.01)	−0.18 (−0.34, −0.03)	−0.12 (−0.17, −0.06)	−0.04 (−0.07, −0.01)	−0.64 (−1.29, 0.01)	−0.07 (−0.18, 0.04)	−0.22 (−0.51, 0.08)	−0.12 (−0.17, −0.06)		
Subgroup analysis of BPA exposure and mature oocytes of IVF/ICSI											
By urinary dilution adjustment method (β (95 % CI)		By BPA level (β (95 % CI)			By country (β (95 % CI)						
SG-adjusted	Unadjusted	$\leq$ 1.70 (ng/ml)		> 1.70 (ng/ml)	China	Poland	America				
−0.05 (−0.08, −0.03)	0.03 (−0.04, 0.10)	−0.03 (−0.06, −0.00)		−0.07 (−0.12, −0.02)	−0.03 (−0.06, −0.00)	−0.22 (−0.51, 0.08)	−0.07 (−0.12, −0.02)				
Subgroup analysis of BPA exposure and blastocyst formation of IVF											
By gender (β (95 % CI)		By urinary dilution adjustment method (β (95 % CI)			By BPA level (β (95 % CI)		By country (β (95 % CI)				
Female	Male	SG-adjusted	Cr-adjusted	$\leq$ 0.16 (ng/ml)	> 0.16 (ng/ml)	China	Slovenia				
−0.04 (−0.10, 0.02)	0.01 (−0.05, 0.06)	−0.02 (−0.05, 0.02)	−0.21 (−0.63, 0.20)	−0.02 (−0.05, 0.02)	−0.21 (−0.63, 0.20)	−0.02 (−0.05, 0.02)	−0.21 (−0.63, 0.20)				
Subgroup analysis of BPA exposure and implantation of IVF											
By gender (β (95 % CI)		By urinary dilution adjustment method (β (95 % CI)			By BPA level (β (95 % CI)		By country (β (95 % CI)				
Female	Male	SG-adjusted	Cr-adjusted	$\leq$ 1.38 (ng/ml)	> 1.38 (ng/ml)	China	Poland	America			
0.47 (0.42, 0.51)	0.07 (−0.58, 0.73)	0.47 (0.42, 0.52)	−0.38 (−1.09, 0.33)	−0.03 (−0.14, 0.08)	0.58 (0.53, 0.64)	−0.38 (−1.09, 0.33)	−0.07 (−0.29, 0.14)	0.50 (0.45, 0.54)			
Subgroup analysis of BPA exposure and high quality blastocysts of IVF											
By gender (β (95 % CI)		By urinary dilution adjustment method (β (95 % CI)			By BPA level (β (95 % CI)		By urinary dilution adjustment method (β (95 % CI)				
Female	Male	SG-adjusted	Cr-adjusted	$\leq$ 0.16 (ng/ml)	> 0.16 (ng/ml)	China	Slovenia				
−0.11 (−0.20, −0.02)	0.03 (−0.05, 0.10)	−0.03 (−0.09, 0.03)	−0.15 (−0.67, 0.37)	−0.03 (−0.09, 0.03)	0.15 (−0.67, 0.37)	−0.03 (−0.09, 0.03)	−0.15 (−0.67, 0.37)				
Subgroup analysis of BPA exposure and high quality embryos of IVF/ICSI											
By gender (β (95 % CI)		By country (β (95 % CI)			Slovenia		Korea	Poland	America		
Female	Male	China	Slovenia	Korea	Poland	America					
−0.08 (−0.14, −0.02)	−0.02 (−0.08, 0.04)	−0.06 (−0.11, −0.01)	0.07 (−0.34, 0.49)	−0.11 (−0.40, 0.18)	−0.01 (−0.31, 0.30)	0.42 (0.09, 0.76)					
Subgroup analysis of BPA exposure and normal fertilization of IVF/ICSI											
By gender (β (95 % CI)		By country (β (95 % CI)			Greece		Korea	Poland	America	Slovenia	
Female	Male	China	Greece	Korea	Poland	America	Slovenia				
−0.07 (−0.10, −0.04)	−0.02 (−0.06, 0.01)	−0.04 (−0.06, −0.01)	−0.14 (−0.61, 0.33)	−0.02 (−0.13, 0.08)	−0.01 (−0.41, 0.39)	−0.12 (−0.18, −0.06)	0.06 (−0.24, 0.35)				
Subgroup analysis of BPA exposure and clinical pregnancy of IVF											
By gender (β (95 % CI)		By country (β (95 % CI)			Turkey		Korea	Poland			
Female	Male	China	Turkey	Korea	Poland						
0.00 (−0.07, 0.07)	0.15 (−0.15, 0.44)	−0.30 (−0.71, 0.11)	0.43 (−0.01, 0.86)	0.06 (−0.23, 0.34)	0.03 (−0.07, 0.12)						

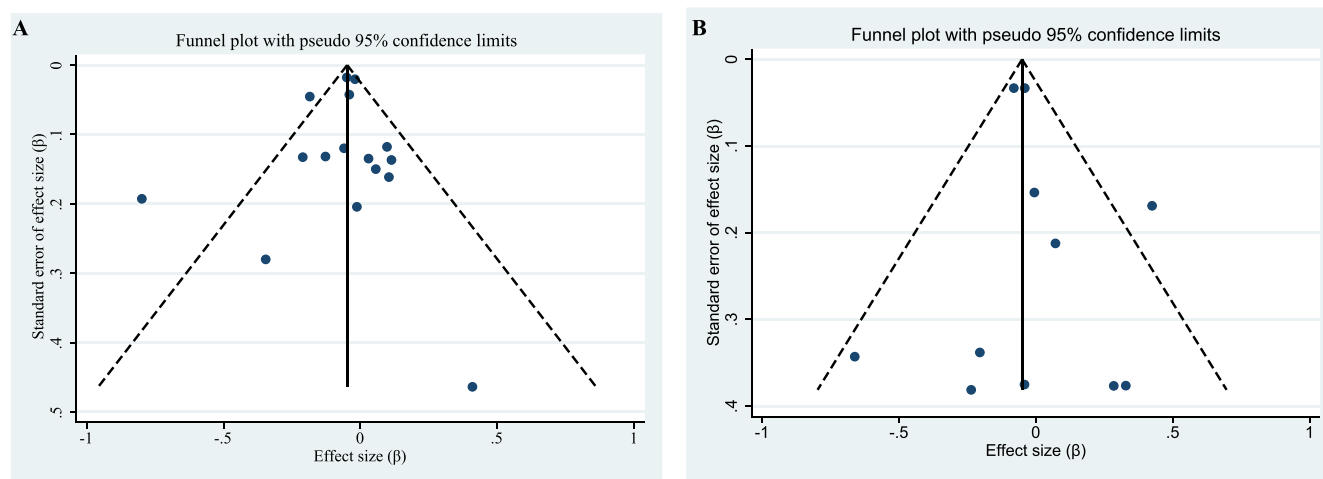


Fig. 5. Funnel plots for publication bias (A) Funnel Plots to detect publication bias for meta-analysis of BPA exposure and normal fertilization of IVF/ICSI. (B) Funnel Plots to detect publication bias for meta-analysis of BPA exposure and high quality embryos of IVF/ICSI. IVF, in vitro fertilization; ICSI, intracytoplasmic sperm injection.

development, which may be attributed to disruptions in maternal-fetal interface immune cells and placental function [39–41]. Lin et al. reported that bisphenol A promotes autophagy in ovarian granulosa cells via the AMPK/mTOR/ULK1 signaling pathway, potentially affecting oocytes, which aligns with the decrease in the number of retrieved and mature oocytes observed in the current study [42]. Therefore, BPA exposure may affect oocyte growth and development by impairing ovarian granulosa cell function, representing a potential pathological mechanism [43,44]. Additionally, BPA exposure, as an endocrine disruptor, can adversely affect pregnancy outcomes such as preterm birth and offspring health, and may also increase transgenerational disease risk through epigenetic modifications [45–47]. The above results indicate that BPA exerts diverse effects on female reproduction and pregnancy, involving endocrine, immune, and epigenetic pathways. However, the biological mechanisms underlying the association between BPA exposure and reduced IVF success remain to be fully elucidated. Overall, the health impacts of BPA are multifaceted and occur through various pathways [48,49]. Considering the widespread exposure to BPA in daily life and its presence in IVF-related equipment, further investigating this issue holds significance as it may adversely affect IVF outcomes. Moreover, our subgroup analysis revealed that high BPA concentration in semen is associated with poor IVF outcomes. Additionally, research has shown that BPA exposure can alter histone epigenetic modifications on spermatozoa, potentially leading to reduced male fertility and adverse IVF outcomes [50,51].

Despite the robust findings regarding fertilization and embryo quality, the limitations of the present study should be acknowledged. First, the heterogeneity observed in the included studies, particularly in exposure assessment methods and population characteristics, limits the generalizability of the results. Furthermore, the majority of the included studies were conducted in America and China, further restricting the generalizability of the results to broader populations. Moreover, observational studies are susceptible to confounding factors, and a causal relationship between BPA exposure and IVF outcomes cannot be established. Finally, this study highlights the association between BPA exposure and intermediate IVF outcomes, but the lack of live birth data in most included studies limits the applicability of our findings to clinical practice. However, the findings should also be interpreted cautiously due to significant methodological limitations, including inconsistent control of confounding factors, challenges in data analysis, and the potential for exogenous contamination and timing-related biases in exposure assessment. Future research should focus on large, multi-center studies with standardized BPA measurement protocols to

confirm our findings and explore the mechanisms underlying BPA's effects on fertility.

In conclusion, this meta-analysis is the first to use large-scale data and highlights the potential risk of BPA exposure on reproductive outcomes in IVF/ICSI. The significant associations between BPA exposure and reduced fertilization rates and embryo quality warrant increased awareness of environmental exposures in reproductive medicine. Considering the widespread use of BPA-containing products, public health measures should be introduced to reduce exposure, especially in individuals seeking fertility treatment. Future research should focus on refining exposure assessment methods and exploring the underlying mechanisms to better inform clinical practices and regulatory policies aimed at protecting reproductive health.

Ethics approval and consent to participate

Not applicable.

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CRediT authorship contribution statement

Cao Xian-ling: Writing – review & editing, Methodology. **Jingyan Song:** Conceptualization. **Xinyan Wang:** Formal analysis. **Zhengao Sun:** Methodology, Data curation. **Mengke Yuan:** Conceptualization. **Jie Chai:** Supervision, Conceptualization.

Authors' contributions

XL conceived the idea for this meta-analysis. All authors (XL, MK, CJ, JY, ZG) developed the methodology for the meta-analysis. The

manuscript was drafted by XL, CJ and MK, and revised by JY, ZG and XL. All authors contributed to the research and agreed to be responsible for all aspects of the work.

Consent for publication

Not applicable.

Declaration of Competing Interest

We declare that there are no conflicts of interest associated with this study.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.reprotox.2025.109071](https://doi.org/10.1016/j.reprotox.2025.109071).

Data availability

Data will be made available on request.

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